

1. ReviewFox: AI-Powered Code Review and Security Analysis Github App and Platform

Introduction

ReviewFox is an intelligent automated code review platform that integrates directly with GitHub to improve software quality, security, and development speed. By combining AI with advanced scanning tools such as Semgrep, SonarQube, and Trivy, ReviewFox provides developers with actionable insights into code vulnerabilities, maintainability issues, and performance problems. The platform reviews pull requests, highlights issues, and offers meaningful recommendations to help teams ship safer, cleaner, and more reliable software.

Statement of the Problem

Engineering teams often face slow, inconsistent, and incomplete code reviews. Overloaded developers, large pull requests, and limited security expertise lead to missed bugs, vulnerabilities, and poor coding practices. Traditional linters and scanners alone fail to provide contextual understanding or practical guidance. As modern applications grow more complex, there is a pressing need for an automated solution that unifies AI reasoning with robust security and quality analysis.

Objectives

- Deliver automated AI-powered pull request reviews with clear, actionable feedback.
- Detects vulnerabilities, insecure coding patterns, and dependency risks early.
- Enhance code quality through Semgrep, SonarQube, and Trivy analysis.
- Improve developer productivity by reducing manual review time.
- Maintain consistent coding standards across all repositories.
- Support junior developers through educational, easy-to-understand explanations.

Significance of the Project

ReviewFox modernizes and strengthens the software development workflow by providing accurate, intelligent, and comprehensive code evaluations. Its hybrid approach—combining AI reasoning with static analysis and vulnerability scanning—ensures deeper insights than any single tool can provide. ReviewFox reduces technical debt, improves security posture,

accelerates development, and contributes to more efficient, collaborative engineering teams. This makes it a valuable and impactful solution for any organization focused on high-quality, secure software delivery.

2. Amazon Clone: Full-Stack E-Commerce Platform

Introduction

The Amazon Clone E-Commerce Platform is a scalable, full-stack online marketplace designed to replicate the core functionalities of a modern e-commerce ecosystem. The platform enables users to browse products, manage shopping carts, securely process payments, track orders, and manage seller accounts. Built with modern web technologies and a robust backend architecture, the system emphasizes performance, security, usability, and scalability.

Statement of the Problem

Many small and medium-sized businesses struggle to establish scalable, secure, and feature-rich online marketplaces. Existing solutions can be expensive, rigid, or lack customization. Additionally, developing a high-performance e-commerce platform requires addressing complex challenges such as secure payments, inventory management, order tracking, fraud prevention, and user experience optimization.

Objectives

- Develop a secure and scalable full-stack e-commerce platform.
- Implement user authentication and role-based access (customer, seller, admin).
- Provide product browsing, filtering, and advanced search functionality.
- Integrate secure payment processing (e.g., Stripe, PayPal, or local payment systems).
- Support order tracking, order history, and delivery management.
- Provide admin tools for monitoring users, orders, and platform activity.
- Ensure responsive design for desktop and mobile devices.

Significance of the Project

The Amazon Clone E-Commerce Platform demonstrates how modern technologies can be leveraged to build a scalable digital marketplace. By integrating secure authentication, payment systems, and efficient database design, the platform provides a real-world simulation of enterprise-level e-commerce systems.

This project serves as both a production-ready marketplace foundation and a comprehensive learning platform for full-stack development, system architecture, and secure transaction handling.

3. Event Booking & Ticketing Platform

Introduction

The Event Booking & Ticketing Platform is a scalable, full-stack web application designed to simplify event creation, ticket sales, and attendee management. The platform enables event organizers to create and manage events, sell tickets online, track sales analytics, and verify attendees using digital tickets with QR codes. Users can browse events, book tickets securely, receive confirmations, and manage their bookings.

Built using modern web technologies and a robust backend architecture, the system emphasizes scalability, security, payment integration, and real-time updates.

Statement of the Problem

Event organizers often face difficulties managing ticket sales, tracking attendance, preventing fraud, and handling payments efficiently. Existing ticketing platforms may charge high service fees, limit customization, or lack proper analytics tools.

Objectives

- Develop a secure and scalable full-stack event management system
- Implement role-based authentication (attendee, organizer, admin)
- Allow organizers to create, edit, and manage events
- Enable users to browse, filter, and search for events
- Generate digital tickets with QR codes
- Implement real-time ticket availability updates
- Develop an admin dashboard for monitoring platform activity

Significance of the Project

The Event Booking & Ticketing Platform demonstrates the practical implementation of a secure, scalable, and feature-rich digital marketplace tailored for event management. By integrating real-time ticket inventory control, secure payment processing, QR-based verification, and role-based access systems, the platform reflects real-world enterprise-level application design. It highlights the ability to architect complex backend logic, manage transactional workflows, and ensure data integrity while delivering a seamless user experience.